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Spatial summation in the tactile sensory system: Probability summation and neural integration

GEORGE A. GESCHEIDER[1,2], BURAK GÜÇLÜ[3], JESSICA L. SEXTON[1], SARAH KARALUNAS[1], & ANNE FONTANA[1]

[1]*Department of Psychology, Hamilton College, Clinton, NY, USA,* [2]*Institute for Sensory Research, Syracuse University, Syracuse, NY, USA, and* [3]*Biomedical Engineering Institute, Bogazici University, Bebek, Istanbul, Turkey*

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Abstract
Psychophysical thresholds for the detection of a 300-Hz burst of vibration applied to the thenar eminence were measured for stimuli applied to the skin through 1.5 cm^2 and through 0.05 cm^2 contactors. Thresholds were approximately 13 dB lower when the area of the contactor was 1.5 cm^2 than when it was 0.05 cm^2. The difference between the thresholds measured with the large and small contactors was significantly reduced when only the lowest thresholds obtained in the testing sessions were considered. This result supports the hypothesis that one component of spatial summation in the P channel is probability summation. In addition, threshold measurements within a session were less variable when measured with the 1.5 cm^2 contactor. We conclude that spatial summation in the P channel is a joint function of two processes that occur as the areal extent of the stimulus increases: *probability summation* in which the probability of exceeding the psychophysical detection threshold increases as the number of receptors of varying sensitivities increases, and *neural integration* in which neural activity originating from separate receptors is combined within the central nervous system rendering the channel more sensitive to the stimulus.

Keywords: *Somatosensation, tactile sensitivity, spatial summation, mechanoreception, neural integration, probability summation, Pacinian corpuscles*

Sensory systems with surfaces containing receptors often become more sensitive as the size of the surface area stimulated increases. This process, referred to as spatial summation, can manifest itself as a decrease in the psychophysically measured detection threshold or, at suprathreshold levels of stimulation, as an increase in the subjective magnitude of the stimulus, and has been studied extensively for tactile stimulation of the skin (Verrillo 1963, 1966a, 1966b; Craig and Sherrick 1969; Franzen 1969; Craig 1976; Gescheider 1976; Bolanowski et al. 1988; Gescheider et al. 1994b, 2002). Four tactile information processing channels are activated when glabrous skin is stimulated—Pacinian (P), non-Pacinian I (NP I), non-Pacinian II (NP II), and non-Pacinian III (NP III). Of the four, only the P channel has been found to be capable of spatial summation of vibratory stimulation (Verrillo 1963; Gescheider 1976; Bolanowski et al. 1988; Gescheider et al. 1994b). For example, when a vibratory stimulus of a frequency high enough to isolate the P channel is applied to a region of the skin containing a significantly dense population of Pacinian corpuscles (PCs), such as the thenar eminence of the hand or the fingertip, the detection threshold decreases as the size of the stimulated skin area increases (Verrillo 1968; Gescheider et al. 2002), thus demonstrating the phenomenon of spatial summation. Little or no spatial summation is observed at lower vibration frequencies that selectively excite NP channels (e.g., Bolanowski et al. 1988; Gescheider et al. 1994b), although using single and double probe stimuli, Greenspan et al. (1997) reported evidence for spatial summation when exciting non-Pacinian nerve fibers.

Prior to this study, a definitive explanation of the phenomenon of spatial summation had not

Correspondence: G. A. Gescheider, Department of Psychology, Hamilton College, Clinton, NY 13323, USA. Tel: +1 315 859 4225. Fax: +1 315 859 4807. E-mail: ggeschei@hamilton.edu

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been provided. Nevertheless, based on our earlier work on temporal summation in the P channel it was our initial hypothesis that spatial summation in this channel results from a combination of two processes—neural integration and probability. *Neural integration* in which neural activity is added up over the time of exposure to the stimulus (e.g., Zwislocki 1960) and *probability summation* in which, as a result of random fluctuations in sensitivity over time, it is more likely that the stimulus will exceed the detection threshold the longer it is presented (e.g., Watson 1979), are known to underlie the phenomenon of vibrotactile temporal summation in which sensitivity improves as the duration of a stimulus increases (Gescheider et al. 1999).

How is it that neural integration and/or probability summation could also be involved in spatial summation? In temporal summation in the tactile system, neural activity is integrated over time periods of up to approximately 800 ms, yielding progressively lower thresholds as stimulus duration increases (Verrillo 1965a, 1965b; Gescheider 1976; Frisina and Gescheider 1977; Gescheider and Joelson 1983; Checkosky and Bolanowski 1992; Gescheider et al. 1994c, 1996). For stimuli greater than 800 ms in duration, it is more likely that the threshold of the system will be exceeded as exposure to the stimulus continues because of probability summation resulting from random fluctuation in sensitivity over time (Gescheider et al. 1999). In spatial summation, neural integration resulting in the adding up of the neural responses of an increasing number of tactile receptors as the areal extent of the stimulus increases should lead to a progressive decrease in the detection threshold. For probability summation to operate in spatial summation the sensitivity of the system must be variable over the area containing the sensory receptors. Accordingly, the likelihood that the most sensitive receptors will be excited should increase as the areal extent of the stimulus increases to include an increasingly larger number of receptors—thereby resulting in a decrease in the detection threshold. Because the amplitude of vibration needed to produce a threshold response in individual PCs can differ by as much as a factor of ten (Mountcastle et al. 1972; Bolanowski and Zwislocki 1984), it would seem likely that probability summation must be at least a component of spatial summation.

Psychophysical evidence for the operation of probability summation comes from the finding that detection thresholds of the P channel are inversely related to receptor density but only when testing is done with a relatively small contactor (Gescheider et al. 2002). In terms of probability summation, it is only when the area of stimulation is relatively small that the probability of activating highly sensitive receptors is greater at a site of higher than at a site of lower receptor density. When the area of stimulation is large, a sufficiently large sample of the receptor population is encompassed under the contactor to ensure that the most sensitive receptors will be stimulated at both low receptor density and at high receptor density sites. The important psychophysical questions remaining are (a) the extent to which probability summation operates in spatial summation and (b) whether neural integration is also operative after probability summation has been taken into account.

To answer these questions, an approach was used similar to that used by Green and Zaharchuk (2001) in which the components of neural integration and probability summation were successfully isolated in examining the spatial summation of warm and cold sensations. They found that when probability summation was eliminated as a factor in spatial summation, the effect of changing the size of the stimulus on sensations of warmth and cold was reduced. The basic objective of our study, as in that of Green and Zaharchuk (2001), was to determine how thresholds change as a function of the size of the stimulated area under conditions in which the effects of local variations in sensitivity—the underlying basis of probability summation—are eliminated. This was done by sampling thresholds over an area of the thenar eminence of the hand and determining the degree to which thresholds measured at the most sensitive site decreased as the size of the contactor was increased. Indeed, if spatial summation results entirely from probability summation, then detection thresholds measured at the most sensitive sites should not be affected by changing the size of the contactor because these thresholds would always be mediated by the most sensitive receptors. Alternatively, if neural integration contributes to spatial summation, then thresholds measured at the most sensitive sites will be found to decrease as the contactor size is increased because more receptors will be stimulated with the large contactor with their activity integrated at some central nervous system (CNS) site.

Our study, dealing with spatial summation in the tactile system, is based on the fundamental principle, cited by Green and Zaharchuk (2001) but originally established by Hardy and Oppel (1937), that spatial summation cannot be meaningfully interpreted unless local variations in sensitivity are taken into account. In the first experiment, local variations in sensitivity were determined by measuring detection thresholds as the position of the contactor on the skin was varied. In the second experiment, variations in sensitivity over time were determined by measuring thresholds repeatedly when the stimulus was presented at the same site. In the third experiment, psychometric functions were determined for stimuli applied to the same site on the skin with contactors that varied in size.

Experiment I: The effect of local variation in sensitivity of the skin on spatial summation

In Experiment I, local variations in sensitivity of the P channel within a $3.0\,cm^2$ area on the thenar eminence were determined by measuring thresholds for detecting a 300-Hz vibratory stimulus at various sites within the test area. Since the contactors were randomly positioned within the test area, local variations in sensitivity could be assessed using both a large ($1.5\,cm^2$) and a small ($0.05\,cm^2$) contactor. It was expected that, because of spatial summation, the average threshold would be lower for the larger than for the smaller contactor. Also, if probability summation is a component of spatial summation, then the variability in thresholds should be smaller for the larger than for the smaller contactor. Specifically, the theory of probability summation states that, given local variations in sensitivity, the probability of stimulating the most sensitive sites within a testing area should increase as the size of the stimulated area increases. This would result in lower and less variable thresholds. If spatial summation results entirely from probability summation, then the lowest thresholds measured at the most sensitive sites should be the same for the large and small contactors because in both cases the most sensitive receptors would be stimulated. However, if neural integration is a component of spatial summation, then the lowest thresholds measured with the two contactors should be lower for the larger than for the smaller contactor because stimulation with the larger contactor would activate a larger number of receptors, the neural responses of which would be pooled within the CNS.

Method

Four observers, two males and two females, participated in the experiment. The observers ranged in age from 20 to 22 years and all were healthy with no indications of neurological disorders. Prior to the experiment, the observer participated in two or three 1-h practice sessions in which they detected vibratory stimuli. The observer and stimulus-delivering apparatus were located within a sound- and vibration-proofed testing chamber. Vibratory stimuli were produced by a Ling 203-A shaker. Vibratory displacements of the skin were produced relative to a static indentation of the vibrator contactor 0.5 mm into the skin. This static indentation was sufficient to prevent decoupling of the vibrator contactor from the skin during vibratory stimulation (see Cohen et al. 1999). Sinusoidal displacements of the contactor were measured with a calibrated electromagnetic linear–variable displacement transducer (Schaevitz LVDT) mounted on the moving element of the vibrator. To ensure accuracy, all measurements were made with the observer's hand in the same position that it had been in during the time in which psychophysical responses were made. Since the mechanical impedance of the vibrator is very high relative to that of the skin, measurements of the displacement of the vibrating contactor provided direct measurements of the amplitudes of movement of the surface of the skin directly under the contactor during stimulation. The skin of the thenar eminence was stimulated by either a 0.05 or a $1.5\,cm^2$ circular contactor mounted on the moving piston of the vibrator. The edge of the contactor was separated from a rigid surround by a 1.0-mm gap.

In the study of spatial summation in the P channel, it is essential that the biomechanics of the stimulus, as it is conducted from the surface of the skin through epidermal and dermal tissue to PCs, be effectively controlled so that changes in the detection threshold as the size of the contactor is changed can be attributed to changes in the areal extent of the stimulus at the level of the receptors. Two procedures were employed to ensure that this was the case in the present study. The use of the rigid surround to confine the stimulus on the skin surface to the area of the contactor (Verrillo et al. 1983) and the use of a constant static indentation of the contactor to maintain a fixed average distance between the vibrating contactor and the receptors is an attempt to ensure that as the size of the contactor is increased a proportionately larger number of PCs are exposed to the stimulus. Without the rigid surround the vibratory stimulus travels extensively over the surface of the skin, and as a result, it becomes difficult to accurately specify and control the areal extent of the stimulus (Gescheider et al. 1978). Without a constant static indentation of the contactor, the distance between the vibrating contactor and the receptors and the resulting amplitude of displacement of the receptors could be different for contactors of different sizes even though the stimulus displacement at the contactors themselves was the same. Such confounding of contactor size and distance between the contactor and the receptors is exactly what happens when, instead of holding static indentation of the contactor constant, the force of the contactor applied to the skin is held constant (Craig and Sherrick 1969). When force is held constant, pressure increases as the size of the contactor decreases and as a result, static indentation is greater for small than for large contactors. Because increasing static indentation lowers psychophysical thresholds (Makous et al. 1996), it is not surprising that the effects of changing the size of the contactor on the psychophysical threshold are greatly attenuated when the force of the contactor rather than its static indentation is held constant (see Craig and Sherrick 1969). Theoretically, when contactor force is held constant, enhanced sensitivity resulting from exposing more receptors to the stimulus as the size of the contactor increases is partially cancelled by the effects of increasing the distance between the contactor and the receptors. It is for this reason that in our study of spatial summation the stimulus was applied with a contactor having a constant static indentation into the skin to maintain a constant distance between the contactor and the receptors.

Because skin temperature can affect vibrotactile sensitivity (Bolanowski and Verrillo 1982; Verrillo and Bolanowski 1986), skin temperature was held constant to within ±0.5°C of 30°C by a device that circulates water of the appropriate temperature through the hollow chamber of the surround. A Lauda/Brinkman RM-6 heating and refrigeration unit and pump controlled both the water temperature and the circulation flow rate. The observers wore earphones through which narrow-band noise was delivered to mask the sound of the vibrator. Stimulus waveforms and timing were controlled by a Macintosh II computer system. The 300-Hz test stimulus was 1.0 s in duration measured at the half-power point and had a rise–fall time of 10 ms.

Detection thresholds were measured by a two-alternative forced-choice tracking procedure (Zwislocki and Relkin 2001) in which the observer was presented with two sequentially presented observation intervals separated by 1.0 s designated by lights, one containing a stimulus and one not. The presentation of the stimulus was distributed randomly between the two observation intervals with a probability of 0.5. The observer's task was to indicate by pressing one of two buttons which observation interval contained the stimulus. The next pair of observation intervals was presented 1.0 s after the observer responded. The amplitude of the stimulus was decreased by 1.0 dB for every three correct responses (not necessarily consecutive) and was increased by 1.0 dB for every error. This method determines the amplitude of the stimulus resulting in 75% correct responses at the detection threshold. Stimulus amplitude is recorded when performance of the observer is maintained at this criterion and the variability in the tracking record does not exceed 2.0 dB above or below the mean for at least 30 responses. Generally, the observer's tracking would become stable for the required 30 trials after approximately 10–20 responses needed to bring the stimulus amplitude to a level near threshold. Thresholds were expressed in dB referenced to 1.0 μm peak displacement amplitudes. Within a session, five thresholds were measured when the stimulus was delivered through either the 0.05 or the 1.5 cm^2 contactor. It took approximately 5.0 min to measure each threshold after which a period of approximately 3.0 min elapsed during which the contactor was moved to a new position for measuring another threshold. The position of the contactor was randomly varied within a 3.0 cm^2 area on the thenar eminence and location of this testing site was the same for the 1.5 and the 0.05 cm^2 contactor. This was done by repositioning the hand on a grid pattern that was placed on the surface of the table upon which the observer's hand rested during the testing. Five testing sessions were conducted using each of the two contactors.

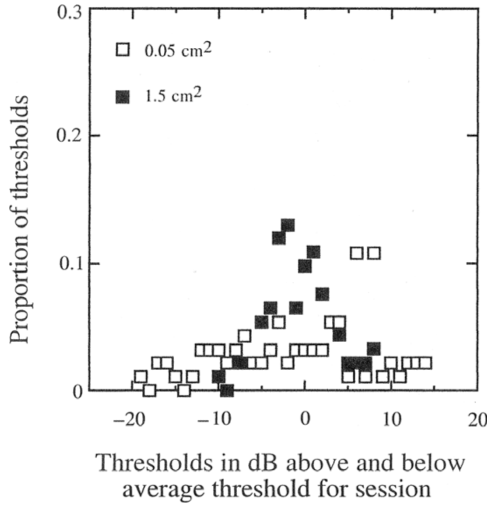

Figure 1. Probability distributions of thresholds measured with a 0.05 cm^2 contactor and with a 1.5 cm^2 contactor when the spatial position of the contactor was varied. Thresholds are expressed as the number of decibels that each threshold was above or below the mean of the five thresholds measured within each experimental session. Frequencies were determined for 0.5 dB steps above (+) and below (−) the mean threshold.

Results and discussion

Plotted in Figure 1 are the distributions of thresholds obtained with the 0.05 and the 1.5 cm^2 contactors expressed in decibels above (positive values) or below (negative values) the average threshold for a testing session. Because all thresholds are expressed relative to the average threshold measured within a session, variability in sensitivity across sessions was eliminated from the analysis as was variability across observers. The primary interest was the variability in sensitivity that could be attributed to moving the contactor from one locus on the skin to another within a 3.0 cm^2 area on the thenar eminence. These distributions were constructed by computing the mean of the five thresholds obtained in a session for the observer and expressing each threshold as a difference in dB from that mean. Counts of the number of thresholds falling within 0.5 dB steps above (+) and below (−) the mean threshold for the session were obtained over the five sessions of the four observers and these were expressed as proportions. The results represent an approximation of how thresholds would have distributed themselves around the average thresholds obtained with each contactor had it been possible to measure a vast number of thresholds within a single session for an ideal average observer. Neither distribution of thresholds deviated significantly from being Gaussian, as indicated by the Kolmogorov–Smirnov test. Consistent with the hypothesis that sensitivity varies substantially within the 3.0 cm^2 testing site and that probability summation operates in spatial summation is the finding that the within-session variability of thresholds measured with the small contactor was substantially greater than that obtained with the large contactor. Indeed, the average within-session standard deviation of measurements was significantly greater ($F[1,3]=124.2$, $p<0.01$) and more than twice as large for the small (3.85 dB) than for the large (1.83 dB) contactor.

From the distributions presented in Figure 1 we constructed cumulative distributions of thresholds that reflect both the variability and the intensity requirements of thresholds measured within a session with each of the two contactors (Figure 2). This was done by adding the threshold values expressed as deviations from the mean threshold in a testing session seen in Figure 1 to the grand mean of thresholds expressed as dB re 1.0 μm peak displacement of the skin calculated over the five sessions and the four observers (−13.53 dB for the 1.5 cm^2 contactor and −0.64 dB for the 0.05 cm^2 contactor). It was then possible to construct cumulative distributions of thresholds in which the proportion of thresholds at or below a particular value of stimulus intensity is plotted as a function of stimulus intensity. The solid curves through the points of the distributions of thresholds represent the best-fitting cumulative Gaussian functions obtained from least squares fits to *z*-score transformations of the proportions.

Examination of Figure 2 reveals that the slopes of these cumulative distributions are different, which means that the variances of the threshold distributions are different for the two contactor sizes. The relatively high variance of the distribution of thresholds obtained with the small contactor indicates that sensitivity of the P channel is far from uniform within an area of 3.0 cm^2 on the thenar eminence. This pronounced variance in sensitivity provides both a basis for and evidence for the operation of probability summation in spatial summation. Specifically, average thresholds were lower when measured with the large (−13.58 dB) than with the small contactor (−0.64 dB) because of the greater likelihood of stimulating the most sensitive spots when the large contactor is used.

The hypothesis that probability summation operates in spatial summation is also supported by the finding illustrated in Figure 3 that spatial summation defined as the decrease in the detection threshold as the size of the contactor is increased is greater for average than for lowest thresholds ($F[1,3]=64.41$, $p<0.01$). However, it is apparent that probability summation cannot account entirely for spatial summation. If it could, then there would be no difference between lowest thresholds measured with the large and small contactors. Indeed, as seen in Figure 2, the lower tails of the distributions of threshold obtained with the large and small contactors do not coincide at the same intensity level. Thus, even when the most sensitive spots were stimulated, thresholds were substantially higher when measured with the small than with the large contactor. If spatial summation had resulted entirely from probability summation, then the lowest thresholds measured with the two contactors would have been the same. But thresholds measured with the large 1.5 cm^2 contactor, capable of exciting 10–20 PCs (Johansson and Vallbo 1979), were lower than almost all thresholds measured with the small contactor able to excite only a few of these receptors.

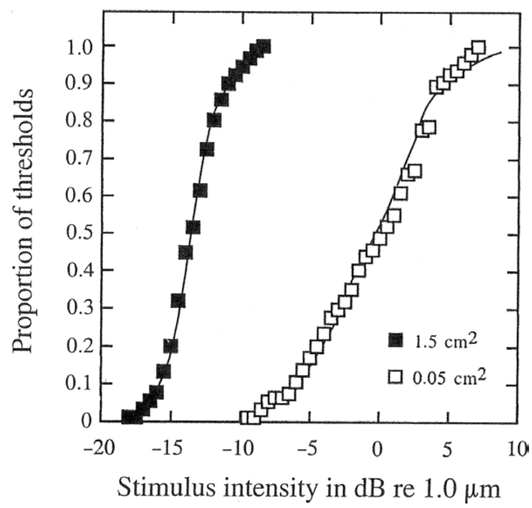

Figure 2. Cumulative probability distributions as derived from distributions shown in Figure 1 of thresholds measured with the two contactors when the spatial position of the contactor was varied. The solid curves are the best-fitting ogive functions (cumulative Gaussian functions) determined by fitting linear functions to *z*-score transformations of the cumulative probabilities.

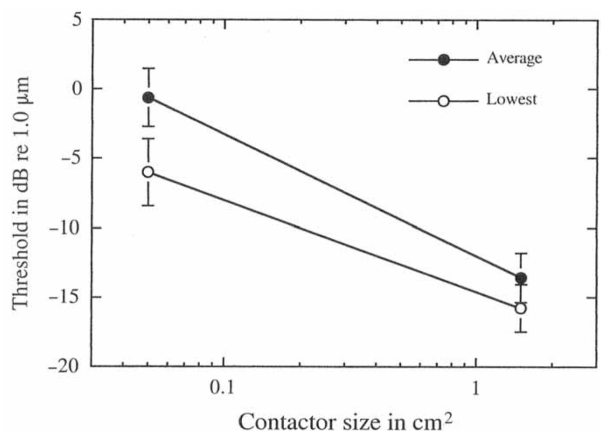

Figure 3. Average thresholds and average lowest thresholds within a session plotted as a function of contactor size.

Thus, the difference in the lowest thresholds measured with the large and small contactors can be interpreted as evidence that the P channel is capable of integrating neural activity over many receptors in achieving improved sensitivity as the size of the contactor is increased.

Examination of Figure 2 indicates that the amount of spatial summation resulting from increasing the stimulus area from 0.05 to 1.5 cm^2 attributable to a process other than probability summation is approximately 8.0 dB, a value corresponding to the difference between the very lowest thresholds obtained with the two contactors. The advantage of using a large contactor to include highly sensitive spots is eliminated when only thresholds measured on the most sensitive skin areas are considered. The remaining 8.0 dB of spatial summation that we have found can be attributed to neural integration.

Fundamental to the concept of neural integration is the premise that activity in the individual peripheral nerve fibers constitutes the input to a neural integrator and that the magnitude of the neural activity of the integrator is a joint function of the number of active fibers providing input and the activity level of the individual fibers. When the activity level of the integrator exceeds threshold, the stimulus is detected. Accordingly, to be detected a stimulus must be presented at a higher intensity, generating higher levels of activity in individual fibers, when it is delivered to a small skin area containing one or only a few receptors than when it is presented to a larger area containing many receptors. Theoretically, the output of the neural integrator and its threshold are the same in both cases, and thus the more intense small stimulus and the weaker large stimulus are equally detectable. It follows from this theoretical conceptualization of neural integration that the intensity of a stimulus required to produce a single spike in a single PC nerve fiber should necessarily be substantially lower than the psychophysical threshold for detecting activity from this same fiber. It is theoretically possible that a weak stimulus capable of initiating single spikes in each of several fibers could be detected, provided the resulting output of the neural integrator exceeds the observer's detection threshold. But if this same weak stimulus were to be applied through a smaller contactor to an area of the skin containing only one receptor, the resulting spike on the receptor's fiber would not constitute a sufficient input to the neural integrator to result in an output above the detection threshold. Only by increasing the level of activity in the input fiber by applying a more intense stimulus could the output of the integrator exceed the detection threshold.

Assuming that the psychophysical thresholds measured with the 0.05 cm^2 contactor approximate the psychophysical detection thresholds when a single PC fiber is stimulated, we can compare these values with those obtained by Mountcastle et al. (1972) for recording a single spike from individual PC fibers. This comparison is presented in Figure 4 as cumulative distributions of neural and psychophysical thresholds. Consistent with the theory of neural integration, the distribution of neural thresholds is substantially lower than that of the psychophysical thresholds. This implies that to be detected, neural activity in one, or at most a few PC fibers, must be substantially higher than a single spike.

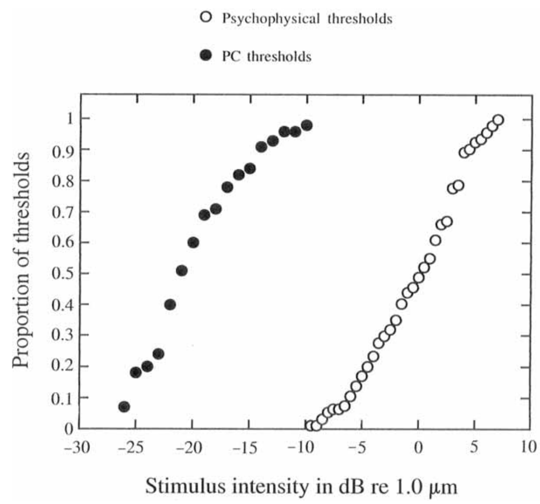

Figure 4. Cumulative distributions of PC thresholds (Mountcastle et al. 1972) and psychophysical thresholds obtained with the 0.05 cm^2 contactor.

Thus, it is our conclusion that at least two fundamental processes—*probability summation* resulting from local variations in receptor sensitivity within an area of skin and *neural integration* within the CNS resulting from the summation of neural activity generated in sensory neurons serving that area—are involved in spatial summation.

Experiment II: The role of variation in sensitivity over time on spatial summation

In Experiment I, it was determined that both probability summation over space and neural integration play a role in spatial summation. In addition to probability summation over space, in which the probability of exciting the most sensitive receptors increases as the size of the stimulated area increases, there is potentially another form of probability summation in which summation can occur over time. As with probability summation over space, probability summation over time is dependent on variability in the sensitivity of the sensory system, but in this case the variability in sensitivity must occur over time rather than over the areal extent of the stimulated receptive surface. We have seen that such variability in sensitivity over time can play a role in temporal summation (Gescheider et al. 1999). The question addressed in this experiment is whether probability summation over time can play a role in spatial as well as in temporal summation.

In order for temporal variation in sensitivity to contribute to spatial summation the thresholds of individual receptors would have to vary over time independently. If the thresholds of individual receptors varied together, then the difference in psychophysical thresholds measured with small and large contactors would not be affected by such variability but the threshold difference would be determined instead by other processes, such as probability summation and neural integration over the areal extent of the stimulated skin area. In contrast, if the sensitivities of individual receptors have some degree of statistical independence in which changes in thresholds over time are not perfectly correlated, probability summation over time could become one of the determinants of spatial summation.

It was predicted that psychophysical detection thresholds measured at the same site on the skin would be lower and less variable within the testing session when measured with a large than when measured with a small contactor. Thresholds obtained with the large contactor should be nearly the same within a 45-min testing period because, although the sensitivities of individual PC (approximately 20 under the 1.5 cm^2 contactor [Johansson and Vallbo 1979]) may fluctuate substantially, there is a high probability that at least some of them will be in their most sensitive state when the stimulus is presented. Because only a single receptor or two would be under the 0.05 cm^2 contactor, the sensitivity of the site should fluctuate considerably from trial to trial as the sensitivity of the receptor(s) fluctuate. This would cause an increase in the variability of the psychophysical thresholds measured within a given session. It was also predicted that if probability summation is controlled over time by considering only the lowest psychophysical thresholds in the threshold distributions obtained with large and small contactors, then the difference between these thresholds should be diminished.

Method

The same four observers who participated in Experiment I also participated in Experiment II. The apparatus was essentially the same as that used in Experiment I. The procedure for measuring threshold was the same as used in Experiment I with the exception that within a session all thresholds were measured at the same site. The observer's hand remained in the same position for the entire session. Before each threshold measurement, the height of the contactor was adjusted to a static indentation into the skin of 0.5 mm. The point at which the contactor barely touched the skin, as determined by the moment at which the electrical resistance between the metal contactor and the skin changed from a reading of open to closed circuit, did not vary from trial to trial within a session by more than ± 0.05 mm. Thus, any variability in thresholds over the session can be mainly attributed to variability within the observer rather than to variability in the stimulus. Five thresholds were measured within each of five testing sessions.

Results and discussion

As in Experiment I, the average threshold for the large contactor (-11.18 dB) was significantly lower than that obtained with the small contactor (1.65 dB) ($F[1,3] = 585.1$, $p < 0.001$), indicating that the amount of spatial summation was substantial (12.83 dB) and nearly identical to the 12.89 dB threshold difference observed in Experiment I.

Furthermore, average thresholds obtained in this experiment were not significantly different than those obtained in Experiment I for these same observers ($F[1,3] = 5.54$, $p > 0.05$).

Plotted in Figure 5 are the distributions of thresholds obtained with the 0.05 and the 1.5 cm^2 contactor. As in Experiment I, all thresholds in Figure 5 are expressed relative to the average threshold within a session, therefore, variability across sessions and across observers was eliminated from the analysis. Variability in sensitivity at the same site within a session attributable to adaptation or practice was not a factor as indicated by the finding that average thresholds measured within a session did not change significantly over the session

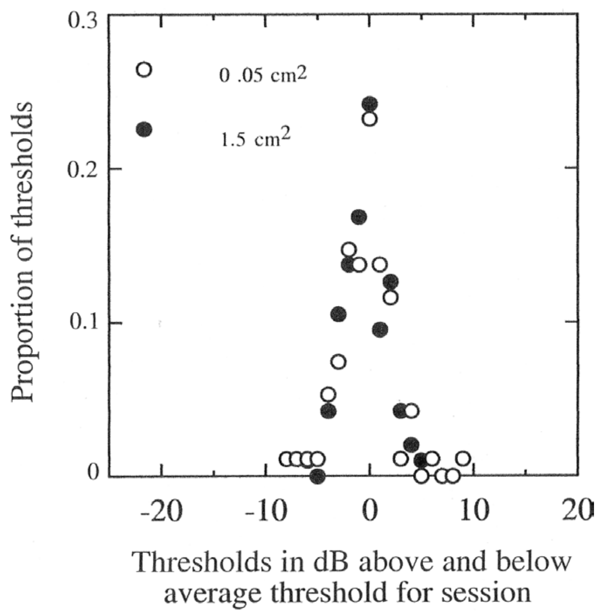

Figure 5. Probability distributions of thresholds measured with a 0.05 cm^2 contactor and with a 1.5 cm^2 contactor when thresholds within a session were measured repeatedly at the same site and frequencies determined for 0.5 dB steps above and below the mean threshold.

($F[4, 16] = 129$, $p > 0.05$). Counts of the number of thresholds falling within 0.5 dB steps above and below the mean threshold for the session were obtained over the five sessions for each observer and over the observers. These counts are expressed as proportions in Figure 5. Although the average difference between thresholds obtained with the large and small contactors was nearly the same in Experiments I and II, the variability in thresholds obtained within a session is very different in the two experiments. When different sites were sampled in Experiment I, and when the same site was stimulated in Experiment II, the average standard deviations of thresholds measured within an experimental session with the large contactors were 1.85 and 0.93 dB, respectively. The smaller standard deviations for thresholds measured in Experiment II must be due to the fact that on successive trials within a session the same sample of receptors is stimulated. The variability in thresholds remaining in Experiment II after the local variability in the sensitivity of sites on the skin is removed can only result from variability in sensitivity over time rather than space. Entirely consistent with this hypothesis is the difference in the variability of the small contactor thresholds in which the standard deviation of thresholds was 3.85 dB when local differences in sensitivity were a factor (Experiment I) but only 1.13 dB when they were not (Experiment II).

In Experiment II one consequence of removing variability in sensitivity within the testing site as a contributor to variability in threshold measurements within a testing session was to render the difference between the standard deviation of measurements with large and small contactors nearly non-existent. The average standard deviations were 1.13 and 0.93 dB for the 0.05 and 1.5 cm^2 contactors, respectively, and the difference between them, though small, was statistically significant ($F[1, 3] = 34.29$, $p < 0.01$). The fact that the distribution of thresholds is slightly less variable for the large than for the small contactor can be interpreted as resulting from the operation of probability summation over time. Although the sensitivity of individual PCs may fluctuate within a testing session, these fluctuations, according to the hypothesis of probability summation over time, are not perfectly correlated among different PCs and consequently their effect will tend to average out and be negligible. In contrast, stimulation through the 0.05 cm^2 contactor involves far fewer receptors, with less chance that fluctuations in sensitivity will be averaged out and as a result, the sensitivity of the site should exhibit greater variability from trial to trial.

From the probability distributions of Figure 5 cumulative probability distributions of thresholds measured within a session with the two contactors were constructed. As in Experiment I, this was done by assigning the average value of the threshold

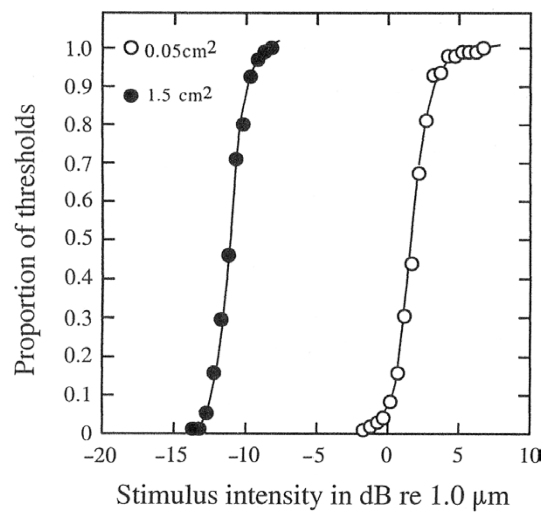

Figure 6. Cumulative probability distributions of thresholds measured with the two contactors when thresholds within a session were repeatedly measured at the same site. The solid curves are the best-fitting ogive functions (cumulative Gaussian functions) determined by fitting linear functions to z-score transformations of the cumulative probabilities.

(−11.18 dB for the 1.5 cm^2 contactor and 1.65 dB for the 0.05 cm^2 contactor) to the mean of the corresponding distribution of Figure 5. As seen in Figure 6, the forms of the distributions were essentially the same as indicated by the fact that the thresholds of both distributions are represented well by the best-fitting cumulative Gaussian functions shown as solid curves. The slightly higher slope of the functions obtained with the 1.5 cm^2 contactor is indicative of the fact that variability in sensitivity as represented by the standard deviation of thresholds measured within a session is inversely related to contactor size. The difference in the variance of thresholds for the two contactors cannot be attributed to changing the site to which the stimulus was applied since this was held constant during the testing session. Instead, the observed difference in variability is consistent with the notion that probability summation over time is one component of threshold variance.

An estimate of the amount of spatial summation uncontaminated by the effects of probability summation over time was made by comparing the lowest thresholds obtained with the two contactors. As seen in Figure 7, the amount of spatial summation expressed as the decrease in threshold when the size of the contactor is increased from 0.05 to 1.5 cm^2 is slightly but significantly less for average lowest thresholds (11.33 dB) than for the average thresholds (12.83 dB) [$F(1, 3) = 10.6$, $p < 0.05$]. Thus, a very small component of spatial summation in the P channel appears to be probability summation over time—small indeed when compared to the larger components of spatial summation—of probability summation over space and neural integration.

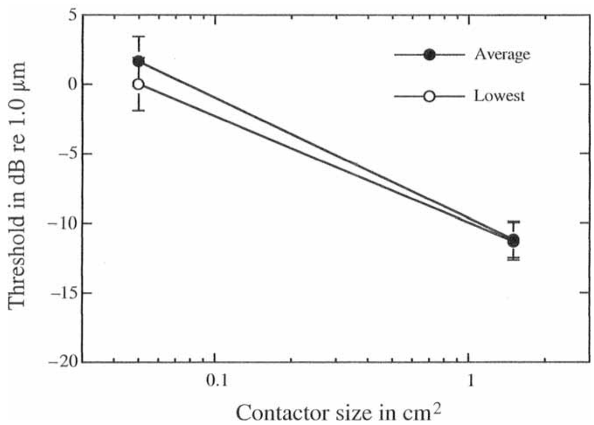

Figure 7. Average thresholds and average lowest thresholds measured at the same site within a session plotted as a function of contactor size.

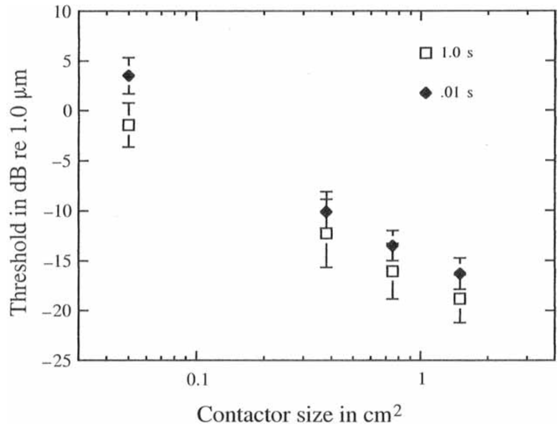

Figure 8. Mean thresholds with standard error bars plotted as a function of contactor size and stimulus duration.

Experiment III: Variability over time as determined by the psychometric function

To test further the hypothesis that a component of spatial summation is probability summation over time, an experiment was conducted in which changes in the probability of detecting the stimulus as a function of stimulus intensity were described as a psychometric function. The slope of the psychometric function, because it reflects variability in sensitivity over time, should be inversely related to the size of the stimulated skin area if probability summation operates in the detection of stimuli (Güçlü et al. 2005). To test this hypothesis, the psychometric function was measured for detection of 300-Hz stimuli delivered through contactors of four different sizes ranging from 0.05 to 1.5 cm^2. In addition, the hypothesis was tested that fluctuations in the sensitivity of the P channel can occur with such rapidity that sensitivity to a 1.0-s stimulus can vary substantially during exposure to that stimulus. These rapid fluctuations in sensitivity make it more likely that the stimulus will be detected but at the same time this reduces trial-to-trial variability, and thereby should steepen the psychometric function. Thus, it was predicted that the slope of the psychometric function should be steeper for a 1.0-s than for a 0.1-s stimulus.

Method

One male and four female observers ranging in age from 20 to 22 years old participated in the experiment. Prior to the experiment, each observer participated in two or three practice sessions. The apparatus was essentially the same as that used in the first two experiments with the exception that contactors with sizes of 0.38 and 0.75 cm^2 were used along with the 0.05 and 1.5 cm^2 contactors. In this experiment, as in Experiment II, all testing within a session was done at a single site. The threshold for detecting a 300-Hz stimulus was first determined by the two-alternative forced-choice tracking procedure used in Experiments I and II. The psychometric function was then determined by presenting nine stimuli, separated in intensity by 1.5 dB, in random order. The nine intensities consisted of two values below the observer's tracked threshold, one at the tracked threshold, and six above the tracked threshold. Over the course of testing, each stimulus was presented 60 times. On each trial, a stimulus was randomly presented in one of two successive observation intervals. The observer was required to report which observation interval contained the stimulus. Psychometric functions, in which the proportion of correct responses is plotted as a function of stimulus intensity, were determined for stimuli of 0.1 and 1.0 s duration. Within a testing session, data were obtained for stimuli delivered through one of the four contactors. The order in which these eight conditions occurred over the course of the experiment was random.

Results and discussion

The mean threshold values obtained by the two-alternative forced-choice tracking are plotted in Figure 8 as a function of contactor size and stimulus duration. Consistent with many previous studies, thresholds decreased as the size of the contactor increased (Verrillo 1963, 1966a, 1966b; Gescheider 1976; Gescheider et al. 2002) and the duration of the stimulus increased (Verrillo 1965a; Gescheider 1976; Gescheider and Joelson 1983; Checkosky and Bolanowski 1992; Gescheider et al. 1994a, 1994b, 1999, 2002). The magnitude of the effects obtained in this study—a decrease in threshold of approximately 3.0 dB per doubling of contactor size and a decrease of 3.0–4.0 dB when stimulus duration was increased from 0.1 to 1.0 s—is entirely consistent with the results of our earlier studies. Thus, it appears that the processes of both spatial and temporal summation were operating in the typical fashion for the observers who participated in Experiment III.

To obtain reliable estimates of the slopes and forms of the psychometric functions for each of the eight contactor size–stimulus duration conditions, the data of the five observers had to be normalized so that they could be combined. The normalization procedure required that for each condition of the experiment the average tracked threshold of the five observers be substituted for the tracked threshold. This value consisted of the third highest value on the averaged psychometric function. The remaining eight intensity values of the psychometric function were 1.5 and 3.0 dB below this value and 1.5, 3.0, 4.5, 6.0, 7.5, and 9.0 dB above this value. The average proportion of correct responses was then computed for each of the nine intensity values of the psychometric functions. Shown in Figure 9 are the psychometric functions determined in this way for the eight conditions of the experiment. The slopes of the functions were quantitatively determined from linear regression fits to the data replotted with proportions converted to *z*-scores (Figure 10). The excellent fits of linear functions to the results indicate that the psychometric functions can be accurately represented as cumulative Gaussian functions.

In Figure 11 the slopes of the linear functions of Figure 10 are plotted against the size of the contactor. As predicted from the hypothesis that a component of spatial summation is probability summation over time, the slope of the psychometric function was directly related to the size of the skin area to which the stimulus was applied. Best-fitting regression lines for each stimulus duration are seen in the figure. Values of r^2 for the relation between log contactor size and slope of the psychometric function were significant at durations of 0.1 s ($r^2 = 0.971$) and 1.0 s ($r^2 = 0.999$). It is also clear that the slopes of the psychometric functions were not attenuated by reducing the duration of the stimulus to 0.1 s. Therefore, the results of this experiment are consistent with the idea that fluctuations in the sensitivity of receptors occur at a relatively slow rate, perhaps taking 1.0 s or more to affect the detection threshold. Gescheider et al. (1999) estimated that fluctuations in sensitivity large enough to result in probability summation over time in temporal summation occur in time intervals that are 0.8 s or longer in duration.

The psychometric functions obtained in this experiment allowed us to consider an alternative biomechanical explanation of spatial summation involving neither neural integration nor probability summation. In this explanation, the detection of the stimulus depends solely on the excitation of some critical number of spatially distributed receptors rather than on the adding up of neural activity from different receptors (neural integration) or on the likelihood of exciting the most sensitive receptors (probability summation). Accordingly, when using a large contactor, the critical number of receptors is excited by a low intensity stimulus, which, as a result of its low amplitude spreads minimally beneath the surface of the skin and thus, for the most part, excites receptors that are directly under the contactor. In contrast, when using a small contactor, the stimulus is detected when its amplitude is increased so that its spread, beyond the area of the contactor, is sufficient to excite the same critical number of receptors needed for stimulus detection with the large contactor. According to this principle, when stimulus amplitude on the surface of the skin is held constant the differential spread at the level of receptors for

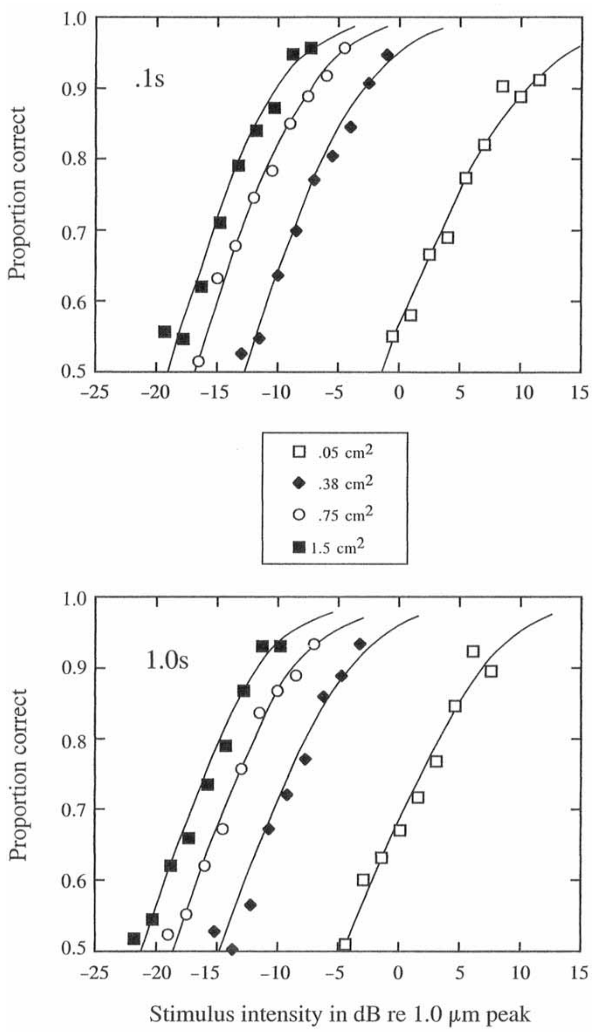

Figure 9. Psychometric functions in which the proportion of correct responses in a two-alternative forced-choice detection task is plotted as a function of stimulus intensity. In the top panel, the data were obtained for stimuli 1 s in duration with contactors varying in size from 0.05 to 1.5 cm^2. In the bottom panel are psychometric functions obtained with the same set of contactors through which stimuli with a duration of 1.0 s were presented.

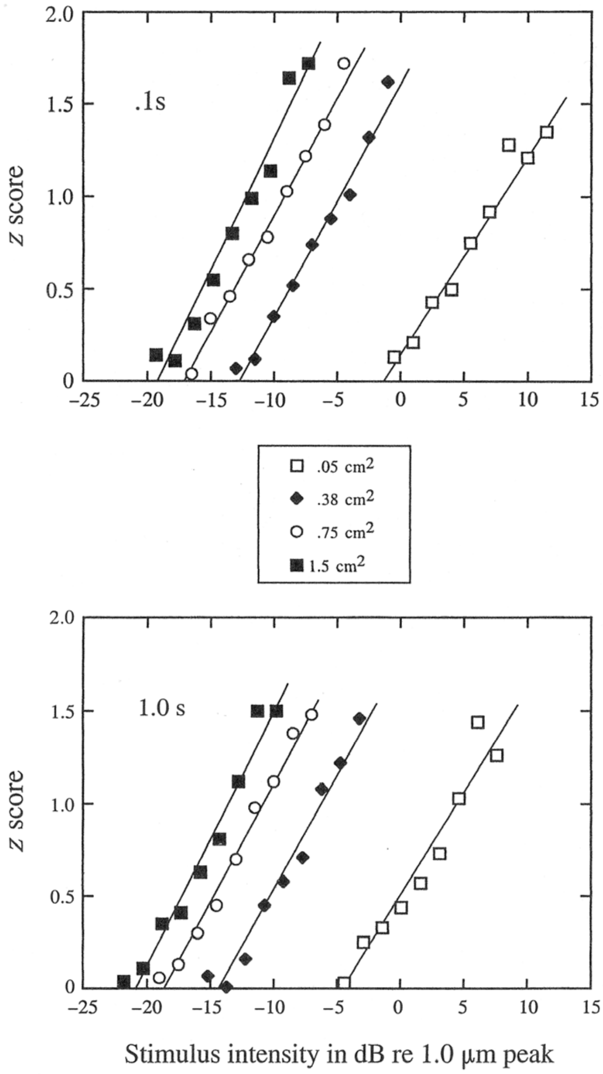

Figure 10. Psychometric functions for the data plotted in Figure 9 with proportions expressed as z-scores. Linear fits to the function were determined by least squares.

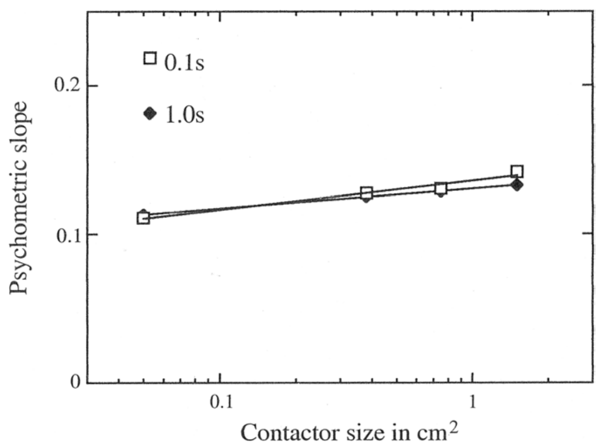

Figure 11. Slopes of z-score psychometric functions for stimulus duration of 0.1 and 1.0 s plotted as a function of contactor size.

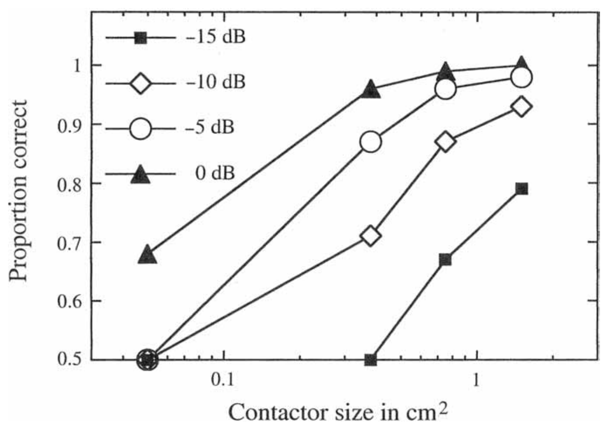

Figure 12. Proportion of correct responses as a function of contactor size at four different constant stimulus intensity levels. Derived from the psychometric functions of Figure 9.

different contactors should be eliminated—since the amount of spread beyond the contactor depends on stimulus amplitude. Consequently, under these conditions, there should be a very dramatic deterioration in the observer's performance in detecting the stimulus as the size of the contactor is reduced and fewer than the critical number of receptors are excited.

Plotted in Figure 12 is the proportion of correct responses as a function of contactor size at various stimulus intensity levels derived from the psychometric functions in Figure 9. At any particular intensity level, performance declines as the size of the contactor decreases. Unfortunately, this finding does not differentiate between the three potential mechanisms of spatial summation–biomechanical effects, neural integration and probability summation. With each of these processes one expects a decline in performance when a stimulus of constant amplitude is applied through the surface of the skin through contactors of decreasing size. Thus, although the results of all three experiments of the present study as well as our mathematical model (Güçlü et al. 2005) are consistent with the hypothesis that probability summation and neural integration are components of spatial summation the possibility that biomechanical factors are also involved cannot yet be ruled out.

Conclusions

An important finding of the present study is that both the variability of and the average threshold measurements within a testing session for detecting a 300-Hz stimulus by the P channel are substantially lower when the stimulus is applied to the thenar eminence at different sites through a large contactor than when it is applied through a small contactor. Indeed, within a $3.0\,\mathrm{cm}^2$ area on the thenar eminence, thresholds of the P channel for detecting a 300-Hz stimulus delivered through a $0.05\,cm^2$ contactor vary over a range of 17 dB, whereas thresholds for detecting the stimulus delivered within the same $3.0\,cm^2$ area through a $1.5\,cm^2$ contactor varied over only a 9.5 dB range. This finding strongly suggests that one component of spatial summation must be probability summation over space, in which the probability of stimulating the most sensitive skin sites within the $3.0\,cm^2$ area increases as the size of the contactor increases. That probability summation resulting from local variability in receptor sensitivity is a component of spatial summation is also clearly indicated by the finding that the difference between thresholds measured with the large and small contactors was significantly reduced when only the lowest, in contrast to all, thresholds measured within a session were considered. The fundamental concept here is that the difference between the average threshold obtained for each contactor is affected by probability summation, in which the probability increases that the most sensitive sites will be stimulated as the size of the contactor increases. Of course, this would also be true when only a single threshold is measured with each contactor and its placement position within a chosen skin area is random. The only circumstance in which probability summation is eliminated is when the lowest thresholds obtained with each contactor size are considered, provided a sufficiently large number of measurements have been made to ensure that the most sensitive sites have been sampled for each contactor size.

Consistent with the hypothesis that the detection threshold is affected by local variations in sensitivity within a larger testing area is the finding that when thresholds were measured repeatedly at the same site, the within-session variability in measurements was substantially lower than when thresholds were measured at different sites—with the range over which thresholds varied decreasing from 9.5 to 5.5 dB and from 17 to 8.5 dB for large and small contactors, respectively. At the same time, the variability in thresholds measured at the same site within a session was lower for the large (range = 5.5 dB) than for the small (range = 8.5 dB) contactor. In addition, as found in Experiment III, the slope of the psychometric function increases, indicating a decrease in the variability of sensitivity, as the size of the stimulated skin area is increased. We posit that when testing repeatedly at the same site, the decrease in the variability of sensitivity as contactor size increases results from probability summation over time, although the effect is small. Assuming that the sensitivities of individual receptors fluctuate independently, there is a high probability that on every trial a subset of them will be in their most sensitive states capable of responding to a weak stimulus, although the particular subset can change from trial to trial. Given this, the psychophysical thresholds should not vary much from trial to trial when the contactor is large enough to stimulate an area of the skin containing a sufficient number of receptors to ensure that the number of receptor activations needed to exceed threshold will occur reliably from one trial to the next. Logically, then, it follows that when the stimulus is applied through a small contactor to an area of the skin containing only a few receptors, the sensitivity of the site will vary greatly from trial to trial as will the psychophysical detection threshold. Constituting further support for the hypothesis that a component of spatial summation is probability summation over time is the fact that spatial summation, expressed as the difference in thresholds measured with the large and small contactors, is reduced when only the lowest thresholds, in contrast to the average thresholds measured at the same site, are considered.

Is probability summation the only underlying factor in spatial summation? If this were true, then the lowest thresholds measured with small and large contactors should have been the same because the detection threshold is determined by the most sensitive sites in both cases. Instead, the lowest thresholds measured with the large contactor were still substantially lower than those measured with the small contactor. Because of this fact we conclude that another factor must be involved in spatial summation. In addition to the probability of stimulating the most sensitive sites as the size of the contactor is increased, is an increase in the sheer number of receptors activated, provided their neural thresholds are exceeded by the stimulus. The underlying process that we presume to mediate the improvement in the psychophysical threshold as the number of activated receptors increases is neural integration, a process probably occurring at some postsynaptic level within the CNS in which neural activity originating from different receptors is combined to produce a larger integrated neural response as the number of activated receptors increases. Implied by this process of neural integration is the idea that the detection threshold in the P channel is determined by the activity of more than a single receptor. Support for this hypothesis is found in the discovery that temporal summation, measured psychophysically as a decrease in the detection threshold as the duration of the stimulus is increased, cannot be accounted for by the neural response of a single PC but instead requires the assumption that the activity of several corpuscles is involved (Checkosky and Bolanowski 1992).

How much of the difference between the average thresholds measured in our study with the 1.5 and $0.05\,cm^2$ contactors can be attributed to neural integration and how much to probability summation? Also, of the amount attributed to probability summation, what portions can be attributed to probability summation over space and to probability summation over time? When the location of the contactor was varied within a testing session, the average difference between thresholds measured with the large and small contactors was about 13.0 dB, but, as estimated from Figure 2, this difference was only 8.0 dB when only the lowest thresholds were considered. If this 8.0 dB difference is a pure measure of the amount of spatial summation attributable to neural integration, it is concluded that of the 13.0 dB difference in average thresholds measured with the two contactors, 8.0 dB of the 13.0 dB results from neural integration while the remaining 5.0 dB results from probability summation. By testing at the same site repeatedly in Experiment II, variability in the sensitivity of different test sites was eliminated as a factor, and differences between the threshold values obtained with the two contactors is only about 1.0 dB less when the lowest rather than the average thresholds are considered. Thus, we conclude that only about 1.0 dB of the 5.0 dB of spatial summation attributed to probability summation results from probability summation over time, whereas the remaining 4.0 dB results from probability summation over space.

The present analysis of spatial summation in the P channel of the tactile sensory system reinforces the fundamental point made many years ago by Hardy and Oppel (1937) that spatial summation cannot be meaningfully interpreted unless local variations in sensitivity are taken into account, which, as we have seen, can result in probability summation over space. In addition, the results of the present study demonstrate that a small component of probability summation can be attributed to probability summation over time. When probability summation is eliminated as a determinant of threshold, it appears that neural integration constitutes the largest contributing factor to spatial summation in the P channel. But, are there biomechanical factors that also contribute to spatial summation? As of now, biomechanical factors have not been ruled out even when a rigid surround to limit the spread of the vibratory stimulus on the surface of the skin and a constant static indentation of the contactor into the skin to control the distance between the contactor and the receptors are used.

The potential for biomedical factors, such as the spread of the vibratory stimulus beneath the surface of the skin, playing a role in spatial summation becomes increasingly important when the distance between the contactor at the surface of the skin and the receptors below is great. Indeed, the PC receptors of the spatially summating P channel are further from the surface of glabrous skin than other cutaneous mechanoreceptors with such distances being as great as 1.0–3.0 mm. It is yet to be determined whether this situation in combination with other physical characteristics of the tissue and properties of the stimulus such as its frequency content provide a biomechanical explanation of some aspects of spatial summation.

Acknowledgements

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